Tuesday, Nov 10 · Bell work and side boards

Today's skill: Distribute (watch the negative) and divide EVERY term

Bell work (at your desk)

1. $-3 - (-8)$
2. $y = -2x + 5$. Find y when $x = -3$.

Level 1 · Integers

a. $-6 + 10$

b. $4 - 9$

c. $-7 - (-3)$

d. $(-5)(-4)$

e. $-18 \div 3$

Level 2 · Distribute

a. $3(x - 4)$

b. $-2(x + 5)$

c. $-4(x - 3)$

d. $\frac{1}{2}(6x - 10)$

Level 3 · Divide every term

a. $(6x + 12) \div 3$

b. $(-10x + 20) \div (-5)$

c. $(-6x - 9) \div (-3)$

d. $-4y = 8x - 12$. Get y alone.

Wednesday, Nov 11 · Bell work and side boards

Today's skill: Undo in reverse order (shoes off before socks) and multiply by the reciprocal

Bell work (at your desk)

1. $8x - 4y = 20$. Get y alone.
2. $5 - (-2)$

Level 1 · Two-step equations

a. $2x + 5 = 17$

b. $3x - 7 = 14$

c. $x/4 + 2 = 5$

d. $-2x + 3 = 11$

Level 2 · Fractions: multiply by the flip

a. $(2/3)x = 8$

b. $(3/4)x = -9$

c. $-(1/2)x = 6$

d. $(5/9)x = 10$

Level 3 · Find b

a. $y = 2x + b$ goes through $(3, 10)$

b. $y = -3x + b$ goes through $(2, 1)$

c. $y = ½x + b$ goes through $(-4, 5)$

d. Solve $y = mx + b$ for b

Thursday, Nov 12 · Bell work and side boards

Today's skill: Number lines: compare negatives, read the four symbols, open vs closed

Bell work (at your desk)

1. V = ℓwh. Solve for h.
2. Which is greater, −7 or −2?

Level 1 · < or >

a. −3 __ 2

b. −8 __ −5

c. −1 __ −10

d. 0 __ −4

Level 2 · Words to symbols

a. at least 5

b. no more than 12

c. more than −2

d. under 30

Level 3 · Is it a solution?

a. $x + 4 > 1$ · try $x = -3$

b. $x + 4 > 1$ · try $x = 0$

c. $2x - 1 \leq 5$ · try $x = 3$

d. $2x - 1 \leq 5$ · try $x = 4$

Monday, Nov 16 · Bell work and side boards

Today's skill: Multiplying and dividing by a negative, and why it flips the order

Bell work (at your desk)

1. $3.75b + 25 \geq 400$
2. $(-6)(-2)$ and $-15 \div (-3)$

Level 1 · Signs

a. $-4 \cdot 3$

b. $-20 \div (-4)$

c. $18 \div (-6)$

d. $(-1)(-9)$

Level 2 · One step with negatives

a. $-3x = 12$

b. $-5x = -35$

c. $x/(-2) = 6$

d. $-x = 8$

Level 3 · Does the sign flip?

a. $3 < 7$ · multiply both by -2

b. $-4 < 8$ · divide both by -4

c. $10 > 5$ · multiply both by -1

d. $2 < 6$ · add -3 to both

Tuesday, Nov 17 · Bell work and side boards

Today's skill: Slope from two points with negatives (subtract in the same order)

Bell work (at your desk)

1. $-2(x - 3)$
2. $5 - 3x > 11$

Level 1 · Slope

a. (1, 2) and (3, 8)

b. (−2, 5) and (2, −3)

c. (−3, 1) and (3, −3)

d. (0, −3) and (5, −3)

Level 2 · Parallel and perpendicular slopes

a. parallel to $y = 4x - 1$

b. perpendicular to $y = 2x + 3$

c. perpendicular to $y = -(3/5)x + 1$

d. perpendicular to $y = x$

Level 3 · Substitute, then plot

a. $g(x) = 2x$ for $x = -1, 0, 1, 2$

b. $h(x) = -x + 3$ for $x = -2, 0, 3$

c. Plot both. Which is steeper?